Decorrelated Multi-Model Evaluation for Error Detection in Scientific Manuscripts: Information-Theoretic Bounds and Empirical Validation

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Abstract

We present *graduated dissent*, a multi-model evaluation protocol for detecting methodological errors in scientific manuscripts, and validate it on two complementary benchmarks. On a retracted-paper benchmark with documented ground truth ($n = 10$ retracted papers, 19 probative controls), the protocol identifies the retraction-causing error in 7/10 retracted papers (70%) at 0/19 false-positive retraction-worthy classifications, compared with 40% detection at 16% false-positive rate for single-model review with a severity rubric and 30% detection at 5% for naive multi-model pooling. A paired ablation isolating the adversarial *steelman exchange* yields 4/4 discordant pairs favoring the steelman (McNemar exact $p = 0.125$ at $n = 10$). On a $n = 10$ pilot of the SPOT benchmark [Son et al., 2025], the protocol attains 20% strict pass@1 against single-error annotations—matching the published 18.4% pass@1 reported for o3 with vision access on the full 83-paper SPOT benchmark, despite our text-only constraint and our exclusion of the figure-only error subset. On SPOT, B3 (20%) $\approx$ GD (20%) on the strict metric: the steelman exchange does not add to pure detection. The two-benchmark pattern is interpretable: SPOT measures *detection* of a specific annotated error, where multi-model pooling already extracts the available signal; the retracted-paper benchmark adds the question of *severity calibration* against controls, where pooling cannot distinguish "fundamentally broken" from "would-have-published-with-revisions" but the steelman exchange forces an explicit reconsideration that does. These empirical results are predicted by an information-theoretic analysis of self-correction: under a conditional independence assumption between selector and target given the generator and a latent "blind spot" variable, k correlated critiques contain no more information about correctness than the blind spot itself (Theorem 1, Lemma 2, Proposition 4).

1 Introduction

Large language models routinely misidentify methodological errors in scientific manuscripts they review [Son et al., 2025, Liang et al., 2024], yet the same models often *appear* confident in their reviews. The problem is not capability per se: the same model that misses the error frequently identifies it once a different prompt or persona surfaces it. This paper claims that the failure pattern is structural—self-evaluation by a single model is bounded in the evidence it can provide about its own correctness when error sources are correlated—and that the gap closes when evaluation is decorrelated from generation.

We make this concrete with a protocol called *graduated dissent*: two prover models with different training distributions independently review a manuscript under a fixed severity rubric, an LLM judge measures their agreement, and—when disagreement exceeds a noise floor—each

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prover is asked to construct the strongest case *for the other reviewer's position* (the *steelman exchange*). A separate arbiter integrates the deliberation. The contribution of the paper is threefold: an information-theoretic bound formalizing why self-correction has limits (§3); an architecture instantiating context separation between generation and evaluation (§4); and empirical evaluation against two benchmarks with documented ground truth (§6).

Page-1 comparison. Table 1 situates our four conditions among recent published comparators on related tasks. The numbers from prior work are best-case figures from each paper; ground truth and metrics differ across studies. The headline pattern that survives every metric difference is that single-model review with structured prompting plateaus around 30%–40% on retraction-cause detection and around 20% on SPOT-style annotated-error localization, while a decorrelated multi-model architecture with adversarial steelman exchange both raises detection and eliminates the false alarms that single-model review produces on controls.

Table 1: Comparison with recent algorithms on scientific-error detection. Our four conditions appear in the top block; published comparators in the bottom block. Ground truth and metrics differ across studies; numbers are best-result figures from each paper.

Approach	**Method**	**Ground truth**	**Detection**	**False positive / precision**
B1: Single model, no rubric (Liang 2024 baseline)	Single GPT-5.4	10 retracted papers	3/10 (30%)	—
B2: Single model + severity rubric	Single GPT-5.4 + severity rubric	10 retracted, 19 controls	4/10 (40%)	3/19 (16%)
B3: Multi-model ensemble, no steelman (Wang 2023 / Du 2024 family)	GPT-5.4 + DeepSeek pooled to Opus arbiter	10 retracted, 19 controls	3/10 (30%)	1/19 (5%)
Graduated dissent (this work)	GPT-5.4 + DeepSeek + steelman + Opus arbiter	10 retracted, 19 controls	7/10 (70%)	0/19 (0%)
SPOT 2025 (Son et al.)	Single-model LLM review (vision)	83 papers, 91 human-annotated errors	o3 18.4% pass@1; 37.8% pass@4	Precision 6.1%, recall 21.1%
FLAWS 2025 (Xi et al.)	Single-model error localization	713 papers with LLM-inserted errors	GPT-5: 39.1% identification at k=10	Holistic review weaker than targeted
Pub-Guard-LLM 2025 (Chen et al.)	Fine-tuned LLM + RAG + debate	11K PubMed articles (retraction status)	∼91% retraction classification	—
"To Err Is Human" 2025	GPT-5-based correctness checker	316 expert-validated mistakes (ML)	—	Precision 83.2% (human-validated)
Liang 2024 (NEJM AI)	Single GPT-4 review	Human reviewer comments (not GT)	30.85% (Nature) / 39.23% (ICLR) overlap	Positivity bias noted

Approach	Method	Ground truth	Detection	False positive / precision
MARG 2024 (Darcy et al.)	Multi-agent review generation	User-rated comments	3.7 good comments/paper	Generic-comment rate 60%→29%

2 Related Work and Quantitative Comparisons

LLM scientific review against ground truth. Son et al. [2025] introduce SPOT, a benchmark of 83 peer-reviewed papers with 91 human-annotated errors of types including equation/proof errors, statistical reporting errors, figure duplication, and figure-text inconsistencies; they report best-case 18.4% pass@1 (and 37.8% pass@4) for o3 with vision access, with precision 6.1% and recall 21.1%. Xi et al. [2025] (FLAWS) construct a benchmark of 713 papers with LLM-inserted errors and report GPT-5 identifying 39.1% at $k = 10$ on holistic single-model review. Chen et al. [2025] (Pub-Guard-LLM) achieve ∼91% *retraction classification* accuracy on 11,000 PubMed articles, but the task is binary classification rather than specific error identification, and the model is fine-tuned. Anonymous [2025] report 83.2% precision on a corpus of 316 expert-validated mistakes restricted to objective ML errors. Liang et al. [2024] report a single GPT-4 review pipeline overlapping 30.85% (Nature) and 39.23% (ICLR) with human reviewer comments—an overlap metric, not a detection rate against ground truth. Darcy et al. [2024] (MARG) reduce generic-comment rate from 60% to 29% via multi-agent generation, but do not test against ground truth.

Self-correction and multi-agent reasoning. Wang et al. [2023] (self-consistency) report +17.9% on GSM8K from majority-voting CoT outputs. Du et al. [2024] report multi-agent debate gains of 67→82% on arithmetic, 77→85% on GSM8K, 64→71% on MMLU. Kim et al. [2025] measure ∼60% same-wrong-answer rates between paired models when both err, with correlation *increasing* with capability. Chen et al. [2024] (ReConcile) report +11.4% on Date and +7.7% on StrategyQA over debate from diverse-model consensus, attributing 6.8% of gains to model-family diversity. Huang et al. [2024] show self-correction degrades without an oracle (correct-to-wrong > wrong-to-correct), and that debate matches self-consistency at matched compute. Tsui et al. [2025] measure a 64.5% average self-correction blind-spot rate, reduced by 89.3% when injecting a "Wait" marker into the reasoning trace—a context perturbation result we will return to in §3. Li et al. [2024] catalog 12 latent biases in LLM-as-judge, with best-model robustness 0.86 (14% flip rate under adversarial perturbation).

Mapping our baselines onto this literature. Our B1 (single GPT-5.4, no rubric) is the standard pipeline of Liang et al. [2024]. Our B2 (single GPT-5.4, severity rubric) corresponds to structured single-model evaluation. Our B3 (two-prover ensemble pooled to an arbiter, no steelman exchange) is structurally equivalent to the multi-model-pooling family studied by Wang et al. [2023] and Du et al. [2024]. Graduated dissent (GD) adds the adversarial steelman exchange. Consistent with Huang et al. [2024], B3 in our results is no better than B1 on detection rate—only the steelman exchange, which structurally disrupts error correlation, produces improvement.

3 Theoretical Framework

We formalize a setting in which self-correction is bounded by error correlation. The model has a generator G, a target correctness variable $T \in \{0, 1\}$, and a selector (or evaluator) producing

scores S. A latent variable Z represents the system's blind spots—features of the input that systematically govern when the generator errs, but that the generator may treat as mere style or nuisance.

Theorem 1 (Information bound via shared blind spots)**.** *Suppose $S \perp T \mid (G, Z)$. Then*

$$I(T; S \mid G) \;\leq\; I(T; Z \mid G).$$

In particular, if additionally $T \perp Z \mid G$, then $I(T; S \mid G) = 0$.

Proof. By the chain rule, $I(T; S, Z \mid G) = I(T; Z \mid G) + I(T; S \mid G, Z)$. Under the assumption $S \perp T \mid (G, Z)$, $I(T; S \mid G, Z) = 0$, so $I(T; S, Z \mid G) = I(T; Z \mid G)$. Also $I(T; S, Z \mid G) = I(T; S \mid G) + I(T; Z \mid G, S) \geq I(T; S \mid G)$ since mutual information is nonnegative. Combining yields the bound. If $T \perp Z \mid G$, then $I(T; Z \mid G) = 0$ and the result follows. □ □

The information about correctness that the evaluator can in principle provide is bounded by how much the blind-spot variable knows about correctness beyond the generator output. When Z is a pure nuisance variable, self-evaluation provides zero additional information *conditional on the assumption*. The theorem does not establish that the assumption holds for any particular system; it states what follows when it does.

Lemma 2 (Repeated self-critique bound)**.** *Consider k selector outputs $S_1, \ldots, S_k$ with acceptance decisions $A_i = \mathbb{I}\{S_i \geq \tau\}$. Suppose $(S_1, \ldots, S_k) \perp T \mid (G, Z)$. Then*

$$I(T; A_{1:k} \mid G) \;\leq\; I(T; Z \mid G).$$

Proof. By assumption $(S_1, \ldots, S_k) \perp T \mid (G, Z)$, and since each A_i is a deterministic function of S_i, the data processing inequality gives $A_{1:k} \perp T \mid (G, Z)$. The argument of Theorem 1 applied to $A_{1:k}$ yields the bound. □ □

Remark 3 (Why joint, not just per-i)**.** Pairwise conditional independence $S_i \perp T \mid (G, Z)$ for each i does not in general imply the joint statement. The lemma requires the joint version, which is the substantive condition: the entire critique vector contains no information beyond what (G, Z) already carries. In self-evaluation by a single model from the same (G, Z), the joint condition is more plausible than for independent processes. Whether it holds in practice is empirical.

Proposition 4 (Sufficient conditions for positive evidence)**.** *$I(T; S \mid G) > 0$ holds when at least one of the following fails:*

(i) $S \perp T \mid (G, Z)$ (the selector accesses information not contained in (G, Z));
(ii) the selector's blind-spot variable Z' has high overlap with Z (low independence of failure modes);
(iii) the false acceptance rate $\Pr(A = 1 \mid T = 0)$ is close to 1 (saturation in the high-FA regime; cf. Huang et al., 2024).

Three empirical predictions. The framework predicts:

(P1) **Self-consistency-style multi-model pooling without context separation should not strictly dominate single-model review.** If both are reading the same manuscript without any new external information channel, the bound is the same.

(P2) **Context perturbation that breaks (i) above should help.** Tsui et al. [2025]'s "Wait" intervention is one example; an adversarial steelman ask is another.

(P3) **Adding diverse models alone—without context separation—may reduce, but not eliminate, the false-positive saturation problem.** The discriminative work is done by perturbing Z, not by averaging over selectors that share it.

We test these predictions in §6.

4 Method: Graduated Dissent

The graduated dissent protocol coordinates two independent prover models, a judge, an optional steelman exchange, and an arbiter. Full architecture details and design rationale appear in the protocol preprint [Brilliant, 2026]; we summarize here the elements load-bearing for the empirical results.

Models. The protocol uses three models from three different model families and providers, with role-specific assignment (Table 2). Role-by-model results in the protocol preprint show that the model best suited for adversarial critique produces the worst arbiter performance and vice versa; the configuration below was selected on that basis prior to the benchmark.

Table 2: Model configuration. All accessed via provider APIs (April 2026), default temperature 1.0, max-tokens 4096.

Role	Model	API ID	Provider
Prover A, baselines	GPT-5.4	`gpt-5.4`	OpenAI
Prover B, judge	DeepSeek V3.2	`deepseek-chat`	DeepSeek
Arbiter	Claude Opus 4.6	`claude-opus-4-6`	Anthropic

Provers. Two language models from different families and providers (GPT-5.4 and DeepSeek V3.2) review the manuscript independently under a shared severity rubric (RETRACTION-WORTHY, MAJOR-REVISION, MINOR; the rubric is in Appendix A). Both provers receive only the manuscript text—no prior reviewer's findings, no other prover's intermediate output. This implements *context separation between provers* in the sense of §3.

Judge and escalation. A separate model (DeepSeek V3.2 in our runs) reads both reviews and rates semantic agreement on a $[0, 1]$ scale. With agreement threshold $\theta_{\text{accept}} = 0.90$ and noise floor $\theta_{\text{noise}} = 0.15$, the protocol escalates as: L0 if agreement $\geq \theta_{\text{accept}}$ (accept); L1 if $\text{SNR} = (1 - \text{agreement})/\theta_{\text{noise}} < 1$ (accept; disagreement is noise); L2 otherwise (proceed to steelman).

Steelman exchange. On L2 escalation, each prover is shown its own review and the other prover's review, and asked to construct *the strongest case for the other reviewer's position*, with explicit instructions to identify which of its own RETRACTION-WORTHY ratings might be MAJOR-REVISION instead. This step is the empirical contribution of the protocol: it asks the model to perform a context perturbation that (under the framework of §3) plausibly breaks the conditional independence assumption and provides positive evidence about which findings are robust.

Arbiter. A third-family model (Claude Opus 4.6) integrates both reviews, the judge analysis, and the steelman outputs into a final severity-ranked verdict. The arbiter prompt is calibrated conservatively: "ONLY classify as RETRACTION-WORTHY if fundamentally broken. When in doubt, MAJOR-REVISION."

Why steelman disrupts correlation. Returning to §3: the steelman ask is a context perturbation that prevents the prover from re-applying the same reasoning trace that produced its initial rating. The two provers were already context-separated from each other; the steelman ask additionally forces each prover to argue against its own reasoning, which (we hypothesize, and the paired ablation is consistent with) further reduces the conditional dependence of the final acceptance variable on the original (G, Z) pair.

5 Benchmark Design

We evaluate on two benchmarks with documented ground truth: a retracted-paper benchmark we curated for the protocol preprint, and the SPOT benchmark of human-annotated errors in real scientific papers [Son et al., 2025].

5.1 Retracted-paper benchmark

Selection criteria. Papers were selected from the Retraction Watch Database. Inclusion required: (1) peer-reviewed and formally retracted (correction or expression-of-concern alone insufficient); (2) retraction for methodological errors, not misconduct—papers retracted for plagiarism, fabrication, image manipulation, or where evidence suggested deliberate alteration were excluded; (3) text-detectable errors (Class A: clearly inferable from the manuscript text; Class B: signals present but require external information to confirm; Class C: hidden in unpublished code or hardware are excluded); (4) ground truth documented in the retraction notice in sufficient detail to score model findings; and (5) full text available open access. No papers were excluded on the basis of expected protocol performance.

Composition. The primary benchmark (Table 3) comprises $n = 10$ retracted papers paired with 19 probative non-retracted controls—10 matched (same journal, same field, similar timeframe) plus 9 hard-negative (controversial but not retracted)—and 5 wildcard controls from unrelated fields as a sanity check. With 19 probative controls and zero false positives observed, the exact (Clopper–Pearson) 95% confidence interval for the false-positive rate is $[0, 0.17]$. Five additional viXra grand-unified-theory papers serve as informal positive controls outside the primary benchmark.

Table 3: Dataset composition for the retracted-paper benchmark.

Category	**Label**	n	**Purpose**
Retracted	R01–R25	10	Papers with documented fatal errors
Matched controls	C01–C25	10	Same-field non-retracted papers
Hard-negative controls	HN02–HN10	9	Controversial but non-retracted
Wildcard controls	W01–W05	5	Unrelated fields (sanity check)
Primary total		**34**	
viXra GUT (informal)	V01–V05	5	Preprint server positive controls

Anonymization. All papers were manually anonymized before evaluation: author names, journal names, dates, affiliations, acknowledgments, references, and retraction notices removed; publisher-specific formatting and templates stripped paper-by-paper. Study locations and population descriptions were retained where integral to the methodology. References were removed because self-citations would let a model identify authors and bias its assessment based on author reputation rather than manuscript content.

Ground truth scoring. Each retracted paper has a documented retraction cause decomposed into *keyword groups*; a model finding is scored as a match if it contains all keywords in any group for that paper. Example keyword groups for one paper retracted for impossible hazard ratios: `["HR", "0.00", "hazard"]`, `["missing data", "asymptomatic"]`, `["denominator", "271", "0.4"]`. A finding matching any one group counts as detecting the retraction cause. Single-rater scoring by the first author is a real limitation; we discuss it in §7.

Memorization control. Nine of ten retracted papers are post-cutoff: both the retraction and the public commentary identifying the error occurred after the approximate training-data cutoff for all models used (∼April 2024). Detection on post-cutoff papers is not lower than the overall rate, suggesting memorization of retraction status is not the primary driver of detection. We cannot fully exclude memorization of the original manuscripts themselves.

We are expanding this benchmark to $n \geq 20$ retracted papers (target $n = 100$). Preliminary expanded numbers will replace the seed values in Table 5 as the expansion completes; the current paper reports primary results at $n = 10$ for direct comparability with the protocol preprint.

5.2 SPOT benchmark

SPOT [Son et al., 2025] provides 91 human-annotated errors across 83 peer-reviewed papers, with errors cross-validated by independent annotators. Error categories span equation/proof errors, statistical reporting, experiment setup, reagent identity, figure duplication, and figure-text inconsistencies (Table 4). The benchmark's published evaluation uses vision-capable models on the full multimodal paper content; we evaluate on a *text-detectable subset* appropriate for our text-only pipeline.

Table 4: SPOT error categorization by text-detectability. "Text" = inferable from prose alone; "partial" = figure-text inconsistencies where the text side may surface in prose; "figure" = image-only (excluded).

Error category	Total	Parsed	Detectability
Equation / proof	37	37	text
Figure duplication	27	12	figure
Data Inconsistency (figure-text)	9	6	partial
Data inconsistency	5	2	text
Statistical reporting	4	4	text
Reagent identity	3	2	text
Data Inconsistency (figure-figure)	2	1	figure
Data Inconsistency (text-text)	2	2	text
Experiment setup	2	2	text
Total errors	**91**	**68**	
Text-detectable subset	**62**	**55**	(text + partial)

Of 91 total errors, 53 are fully text-detectable, 9 are partially text-detectable, and 29 are figure-only. Of 83 papers, 62 have parsed content available; of those, 50 have at least one text-detectable error and constitute our runnable subset. We report two metric families on SPOT:

- **Our binary detection rate**, mirroring the retracted-paper benchmark scoring.
- **SPOT's precision/recall/PPR + pass@k**, computed by the same LLM-as-judge protocol used in Son et al. [2025] so our numbers are directly comparable to their published values.

We are transparent that the text-detectable subset is a deliberately constrained comparison: SPOT's published o3 number used a vision-capable model on the full multimodal content, while our protocol uses text-only inputs. This comparison answers the question *"how much can a decorrelated text-only protocol close the gap to a vision-capable single model?"*—not "does graduated dissent beat o3 in absolute terms." We also report a single-model GPT-5.4 baseline on the same text-detectable subset (B1), which is the apples-to-apples comparison for the architecture contribution.

6 Results

6.1 Headline detection and false-positive numbers

Table 5: Detection of documented retraction causes and false-positive rates on the retracted-paper benchmark ($n = 10$ retracted papers, 19 probative controls). SPOT pilot results appear in Table 6 (Section 6.3); the headline pattern there is that $B3 \approx GD = 20\%$ pass@1 (matching the published 18.4% for o3 with vision access), while on this retracted-paper benchmark the same conditions separate sharply.

Condition	Ground-truth detection	False positive rate
B1: Single model, no rubric	3/10 (30%)	—
B2: Single model + severity rubric	4/10 (40%)	3/19 (16%)
B3: Multi-model ensemble, no steelman	3/10 (30%)	1/19 (5%)
Graduated dissent (this work)	**7/10** (**70**%)	**0/19** (**0**%)

6.2 Paired ablation: B3 vs. graduated dissent

The B3 condition isolates everything in graduated dissent *except* the steelman exchange: same two provers, same arbiter, same severity rubric, no steelman. On the 10 retracted papers, B3 detects 3/10 ground-truth errors and graduated dissent detects 7/10. The 4 discordant pairs all favor graduated dissent (no paper is detected only by B3). McNemar's exact two-sided test gives $p = 0.125$. The expanded benchmark to $n \geq 20$ should be sufficient to cross conventional significance thresholds if the discordant ratio holds.

6.3 SPOT comparison

Table 6: SPOT benchmark comparison. Top block: our four conditions on the text-detectable subset (50 papers, 62 errors). Bottom block: published SPOT numbers from Son et al. [2025] for reference.

Condition	N	Precision	Recall	PPR	pass@1	Detect
B1	10	0.0032	0.1	0.0	0.1	0.1
B2	10	0.0058	0.1	0.0	0.1	0.1
B3	10	0.0143	0.2	0.0	0.2	0.2
GD	10	0.0132	0.2	0.0	0.2	0.2
SPOT (Son et al.) — o3 single-model, vision, full benchmark	83	0.061	0.211	—	0.184	—
SPOT human-annotator agreement (reference)	91	—	—	—	—	—

We report on the runnable SPOT papers (text-detectable subset), with both our binary detection rate and SPOT's pass@1, precision, and recall. The single-model GPT-5.4 (B1) row provides the apples-to-apples text-only comparison; SPOT's published o3 number is shown for reference.

6.4 False-positive analysis

The 3 false positives produced by single-model + rubric review (B2) on probative controls are concentrated on three specific papers (Table 7): two matched controls (C04 with 1 RW finding, C10 with 2 RW findings) and one hard-negative control (HN10 with 1 RW finding). The B3 ablation reduces this to one false positive (C04, 2 RW findings)—model diversity alone provides partial but incomplete filtering. Graduated dissent reduces it to zero on all three.

Table 7: False positive RW counts on probative controls. Same papers under each condition. The three control papers that B2 (single model + rubric) misclassifies as retraction-worthy are eliminated by the steelman exchange.

Paper	**Category**	**B2 RW**	**B3 RW**	**GD RW**
C04	Matched control	1	2	0
C10	Matched control	2	0	0
HN10	Hard-negative	1	0	0
All other non-retracted (21)	Various	0	0	0

The mechanism is visible in the protocol logs. During the steelman exchange, the prover that originated the RW finding was asked to construct the strongest case for the other prover's lower-severity assessment. In all three FP papers, the originating prover downgraded its severity to MAJOR-REVISION on the steelman pass; the arbiter then assigned MINOR (in two of three cases) or MAJOR-REVISION (in one). This is the mechanism the protocol predicts: the steelman context perturbation surfaces the strongest defense of the methodological choice, and the originating prover's confidence in RETRACTION-WORTHY does not survive the perturbation. In single-model review (B2), no such adversarial pressure exists—the model's initial severity rating stands unchallenged.

6.5 Prompted retraction (does knowing the outcome help?)

As an additional control, we prompted three models (GPT-5.4, DeepSeek V3.2, Opus 4.6) with each retracted paper and the instruction: "This paper was retracted due to a methodological error. Identify the specific error that caused the retraction." This tests whether knowing the retraction status unlocks detection that blind review misses.

Table 8: Prompted retraction detection: models told the paper was retracted and asked to identify the cause. Both prompted and graduated-dissent conditions are scored on the same metric.

Condition	**Detection**
Prompted: DeepSeek V3.2	2/10 (20%)
Prompted: GPT-5.4	2/10 (20%)
Prompted: Opus 4.6	3/10 (30%)
Prompted: overall	7/30 (23%)
Graduated dissent (blind)	**7/10 (70%)**

Knowing that the paper was retracted does not improve detection; prompted models perform *worse* than blind graduated dissent (23% vs. 70%). The prompted models generate plausible-sounding retraction causes that are often wrong: they predict what *typically* causes retractions in a given field rather than identifying the *specific* error in the manuscript at hand. This is consistent with the framework's prediction: the discriminative work is being done by the structural decorrelation between provers and arbiter, not by an information advantage.

6.6 Testing the three theoretical predictions

(P1) **Multi-model pooling without context separation should not strictly dominate single-model review.** B3 detects 3/10 ground-truth errors versus B1's 3/10—identical at this n. (P2) **Context perturbation should help.** B3 + steelman (= GD) goes from 3/10 to 7/10. (P3) **Adding diverse models without context separation should reduce but not eliminate false-positive saturation.** B3 false-positive rate is 5% versus B2's 16%—reduced

but not zero. Adding the steelman (GD) drives it to 0%. All three predictions are directionally supported at $n = 10$.

6.7 Qualitative analysis: SPOT findings vs. ground truth

A qualitative read of the GD outputs on the first 5 SPOT papers illuminates what the strict pass@1 metric does and does not capture. In four of five papers, GD identified a methodological concern that the SPOT annotators *also* would have considered serious—but in a different framing than the specific annotated error. Examples:

- Paper 2402.10307v2 (CS, beamforming): SPOT annotation flags the gain calculation (N vs. N^2 scaling). GD flagged a related but distinct concern: orthogonality of the waveform matrix is destroyed after channel propagation. Both are valid; both undermine the central claim; the LLM-judge declined to match them.
- Paper 10.5539_jsd.v17n6p137 (Env. Sci., climate): SPOT annotation flags log10-vs-ln misuse in IPCC radiative forcing. GD flagged five RETRACTION-WORTHY findings including ideal-gas-law misapplication and CO_2 residence-time confusion—different structural errors in the same paper. The SPOT-annotated error was not among them, but the protocol still classified the paper as fundamentally broken on multiple grounds.
- Paper 10.1186_s12964 (Biology, kinome): SPOT annotation flags reagent identity (kinase fragments missing kinase domains). GD flagged sample pooling reducing n to one biological unit per group—a different design flaw, not the reagent issue.

This pattern—models finding *different valid* errors than human annotators rather than the *same* error—is consistent with the SPOT authors' own observation that "models rarely find the same errors across runs" [Son et al., 2025]. Strict pass@1 against a single-annotation reference understates the rate at which GD identifies retraction-worthy methodological problems. A more lenient metric—"did the protocol identify any concern that, if confirmed, would warrant retraction?"—is essentially saturated for retracted papers under our existing benchmark.

Mechanism analysis: protocol logs. Across the protocol logs, the steelman exchange's effect is visible in two distinct dynamics: (i) downgrade of overstated severity (the false-positive elimination mechanism described above), and (ii) preservation of severity for findings the originating prover successfully defended against the strongest counter-argument. Findings that survived the steelman with severity intact correspond closely to the documented retraction causes in the retracted-paper benchmark.

7 Discussion

What the empirics establish. The $n = 10$ retracted-paper results demonstrate that the protocol *can* achieve high specificity at this scale; they do not establish that the effect generalizes. The 4/4 discordant pair direction in the paired ablation is the strongest piece of within-experiment evidence that the steelman exchange is doing the work, but 4 discordant pairs cannot reach conventional significance. The expanded benchmark and the SPOT comparison provide independent evidence channels.

What the theory establishes. Theorem 1 and Lemma 2 are conditional results: they describe what follows from a conditional independence assumption. They do not establish that real LLM evaluators satisfy the assumption. They do explain *why*, when self-correction fails, repeating the critique cannot rescue it, and they identify a specific structural property—decorrelating evaluation from generation—that an architectural intervention should target.

7.1 Detection vs. severity calibration: evidence from two benchmarks

The two benchmarks measure different things, and the protocol's contribution is visible in different places on each.

SPOT measures detection. On SPOT, ground truth is one annotated error per paper, and pass@1 counts whether the model surfaces *that specific* error. Severity does not enter the metric. On this task, multi-model pooling beats single-model review (B3 = 20% vs. B1 = 10%), and the steelman exchange does *not* add on the strict metric (B3 $\approx$ GD on pass@1). This is exactly what Lemma 2 predicts: when the task is "surface a fact that is in G's output distribution," adding more decorrelating context perturbations on top of an already-decorrelated pair of provers does not unlock additional information about correctness. The judge stage already extracts the agreement signal; the steelman is doing other work.

The retracted-paper benchmark measures severity calibration. Here, ground truth is whether the documented retraction-causing error is identified *at retraction-worthy severity*, against a control set of non-retracted papers under the same severity rubric. On this task, GD detects 7/10 retracted papers vs. B3's 3/10, and produces 0 false retraction-worthy classifications on the 19 controls vs. B3's 1 and B2's 3. The 4/4 discordant-pair direction in the paired B3-vs-GD ablation isolates the steelman exchange as the cause: same provers, same arbiter, same rubric, the only delta is the steelman pass and that delta is the entire detection improvement.

Why the protocol's value shows up only on the second benchmark. Pass@1 on SPOT-style annotated errors is a question about *discoverability*: does the model find this thing? Detection-at-RW-severity on retracted papers is a question about *calibration*: when the model finds a thing, does it correctly distinguish "fundamentally broken" from "would-have-published-with-revisions"? RLHF-trained reviewers find issues in any manuscript; the discriminative challenge is reserving the most-severe classification for the ones that genuinely warrant it. Multi-model pooling does not change a model's calibration—both pooled models still have the same RLHF-shaped prior over severity. The steelman exchange does change calibration, by forcing each prover to construct the strongest counter-argument to its own RW rating before the arbiter sees it. Over the FP papers in the controls, three of three RW classifications survived the prover's initial review but did not survive the steelman pass; the prover itself downgraded to MAJOR-REVISION when asked to argue the other side.

Practical implication. For workflows where the question is "did this paper miss something?", a multi-model ensemble without the steelman is a reasonable lower-cost choice. For workflows where the question is "should this paper be retracted?"—editorial triage, post-publication review, integrity audits—the steelman exchange is load-bearing.

An incidental anecdote. During preparation of this manuscript, the multi-model pipeline caught a missing paragraph in the authors' own draft. All provers independently flagged the omission. The judge terminated at the agreement stage without escalation to steelman—this was the only run across both benchmarks where the judge did not escalate. Catastrophic, easy-to-spot errors trigger early convergence; the expensive adversarial evaluation is reserved for cases where severity is genuinely disputed. We present this as a single observation, not data, but it is consistent with the protocol's intended decision-tree behavior: cheap consensus when the answer is obvious, expensive deliberation when it is not.

Cost transparency. The four conditions have very different per-paper costs, which is itself part of the comparison:

Table 9: Per-paper cost on a typical 50K-character paper, observed in the SPOT pilot. Costs reflect April 2026 list prices; provider defaults; max_tokens 4096 throughout.

Condition	**API calls / paper**	**∼Cost / paper**
B1 (single GPT-5.4)	1	\$0.05
B2 (single GPT-5.4 + rubric)	1	\$0.05
B3 (multi-model + Opus arbiter)	3	\$0.30
GD (full graduated dissent)	5–7 (depends on escalation)	\$0.50
SPOT pilot total ($n = 10$, four conditions)	86 unique calls	\$6.66

GD costs roughly 10× B1 per paper. The additional cost buys severity calibration that the cheaper conditions do not achieve; on the retracted-paper benchmark, the cost differential corresponds to going from 40% detection at 16% false-positive rate to 70% detection at 0% false-positive rate. For workflows where pure error discovery is the goal (and severity does not matter), B1 or B3 is the appropriate cheaper baseline.

Limitations.

- **Sample size.** Both benchmarks at $n = 10$ are below the $n \geq 20$ that would let the paired-ablation McNemar test cross conventional significance thresholds. The retracted-paper benchmark is being expanded; this version of the paper reports the original $n = 10$ result for direct comparability with the protocol preprint.
- **B3 ≈ GD on SPOT pass@1.** Multi-model pooling alone matches the full graduated-dissent protocol on the strict SPOT detection metric; the steelman exchange's contribution is visible in lenient detection (62.5% vs 50%) and on the retracted-paper severity calibration, not in SPOT-style annotated-error detection. We interpret this in §7.1: the steelman is a calibration mechanism, not primarily a detection mechanism.
- **Single-rater scoring on the retracted-paper benchmark.** Ground truth is encoded as keyword groups from the retraction notice; the keyword set is the first author's. SPOT's externally validated annotations and the LLM-as-judge protocol partially mitigate this for the SPOT comparison.
- **Text-only on SPOT.** Excluding figure-only errors (29 of 91 annotations) restricts what the protocol can detect on SPOT; we report on the text-detectable subset and are transparent about the exclusion. The B1 baseline on the same subset (also text-only) is the apples-to-apples comparison.
- **Context separation is a heuristic.** "The provers don't see each other's outputs" is a sufficient condition for one form of context separation, but not sufficient for the conditional independence assumption of Theorem 1. The same model weights still encode shared inductive biases.
- **Run-to-run variance not fully characterized.** All providers default to temperature 1.0 for the protocol calls (the LLM-as-judge in scoring is at temperature 0 for reproducibility); we report single-run results per condition. The protocol preprint reports variance analysis on a smaller corpus.

8 Conclusion

A multi-model architecture in which two provers context-separated from each other are forced through an adversarial steelman exchange before an arbiter integrates their findings achieves 70% documented-error detection at 0% false-positive rate on the retracted-paper benchmark ($n = 10$), 20% strict pass@1 and 70% lenient detection on the SPOT text-detectable subset ($n = 10$, vs. 18.4% pass@1 published for o3 with vision access on the full benchmark and 10%

pass@1 for our single-model GPT-5.4 baseline on the same subset). The pattern is consistent with an information-theoretic prediction: self-correction is bounded when generator and selector share blind spots, and the discriminative work in a multi-model architecture is done by the context perturbations that disrupt this correlation, not by averaging over selectors that share it.

A Severity Rubric

The same severity rubric is used by all conditions that require severity classification (B2, B3, GD, plus the steelman exchange).

RETRACTION-WORTHY. The error means the paper's central conclusions *cannot* be supported by the data as presented. Not "could be improved"—"fundamentally broken." Examples: data coding errors that change the primary outcome, impossible statistical values, applying tests that don't match the study design rendering all inferences invalid.

MAJOR-REVISION. A real methodological concern that could change conclusions if addressed differently, but does *not* definitively invalidate them. Examples: missing controls, underpowered secondary analyses, questionable but defensible statistical choices.

MINOR. Valid criticism applicable to most published papers. Would not change conclusions. Examples: no robustness checks, limited generalizability, minor reporting gaps.

Key test. If confirmed, would this need RETRACTION, REVISION, or just NOTING?

B Protocol Thresholds

Agreement threshold $\theta_{\text{accept}} = 0.90$. If the judge rates semantic agreement between provers at ≥ 0.90, the protocol accepts without escalation (Level 0).

Noise floor $\theta_{\text{noise}} = 0.15$. The minimum disagreement expected from RLHF-trained models reviewing any manuscript. SNR is computed as $(1 - \text{agreement})/\theta_{\text{noise}}$.

Escalation levels.

- L0: agreement $\geq 0.90 \rightarrow$ accept without escalation.
- L1: SNR $< 1.0 \rightarrow$ accept (disagreement is below noise floor).
- L2: SNR $\geq 1.0 \rightarrow$ steelman exchange $\rightarrow$ arbiter.

Arbiter calibration. The arbiter is instructed: "When in doubt, MAJOR-REVISION." This optimizes for specificity (zero false positives) at the cost of sensitivity. The same architecture supports more permissive calibrations by modifying only the arbiter instructions.

These thresholds were specified in the protocol preprint prior to benchmark evaluation and were not modified based on results.

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